WORKED SOLUTION

STEP II 1992, Question 13

Mechanics • Friction and Connected Particles

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Read the question before you go on. It is in the STEP Support Programme's question database:

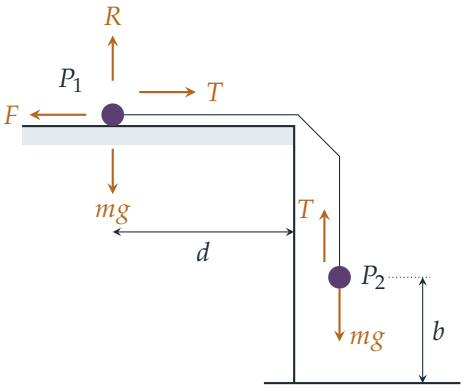
step.maths.org/questions/92-s2-q13

The key idea

The motion has two stages. While P_2 is falling the string is taut and the two particles move together with one common acceleration. Once P_2 hits the floor the string goes slack, and P_1 simply skids to a halt under friction. Everything follows from the speed at the moment of impact: the first part compares a stopping *distance* with the distance left to the edge, and the second compares two *times*.

Solution

We model P_1 and P_2 as particles and the string as light and inextensible, passing over a smooth edge, so the tension T is the same throughout. The table is rough for P_1. We assume $b < d$, so that P_2 lands before P_1 can reach the edge (the string's length plays no further role).



The first part

Stage 1: string taut. Resolving vertically for P_1 gives $R = mg$. It is sliding, so friction is limiting: $F = \mu R = \mu mg$, opposing the motion. With common acceleration a,

$$\begin{aligned} P_2 \text{ (down)}: &\quad mg - T = ma, \\ P_1 \text{ (towards edge)}: &\quad T - \mu mg = ma. \end{aligned}$$

Adding, $2ma = mg(1-\mu)$, so $a = \frac{1}{2}g(1-\mu)$, which is positive because $\mu < 1$; the system really does start to move. P_2 falls a distance b from rest, so its speed V on reaching the floor satisfies

$$V^2 = 2ab = gb(1-\mu).$$

At this moment P_1 has the same speed V, and it has moved a distance b, so it is $d - b$ from the edge.

Stage 2: string slack. After the impact P_2 is on the floor or bouncing up, while P_1 carries on towards the edge, so the string is slack and $T = 0$.

> **Ask yourself** Why can't the string tighten again while P_2 is bouncing?

The string was exactly taut at the moment of impact. After that, P_1 moves closer to the edge, which lets out more string over the edge, and P_2 is no lower than the floor, so it needs no more string than it did at impact. The string stays slack from then on.

So P_1 decelerates at μg alone, and it comes to rest after a further distance

$$s = \frac{V^2}{2\mu g} = \frac{b(1-\mu)}{2\mu},$$

provided that the edge is not reached first. Hence P_1 falls off if and only if $s > d - b$:

$$\begin{aligned} \frac{b(1-\mu)}{2\mu} > d - b &\iff b - b\mu > 2\mu d - 2\mu b \qquad \text{(multiplying by } 2\mu > 0\text{)} \\ &\iff b > \mu(2d - b) \\ &\iff \mu < \frac{b}{2d-b}, \end{aligned}$$

where the last step divides by $2d - b$, which is positive because $d > b$. Every step is reversible, so this is the required "if and only if". (If μ equals $b/(2d-b)$, P_1 stops exactly at the edge and does not fall.)

The second part

Now $\mu > b/(2d-b)$, so P_1 stops on the table. Measure time from the moment P_2 first hits the floor. At that moment both particles have speed V.

P_1: decelerating at μg from speed V, it stops after time

$$t_1 = \frac{V}{\mu g}.$$

P_2: by Newton's law of restitution it rebounds with speed eV. With the string slack it moves freely under gravity, so it returns to the floor when $eVt - \frac{1}{2}gt^2 = 0$, that is, after time

$$t_2 = \frac{2eV}{g}.$$

(The impact is instantaneous, so no time is lost at the first bounce.)

P_1 comes to rest before the second bounce exactly when $t_1 < t_2$:

$$\frac{V}{\mu g} < \frac{2eV}{g} \iff \frac{1}{\mu} < 2e \iff e > \frac{1}{2\mu},$$

dividing by $V/g > 0$. So if $e > 1/(2\mu)$, P_1 comes to rest before P_2 bounces a second time, as required. In fact the converse holds too.

Checks

- Limiting cases: as $\mu \to 0$ the bound $b/(2d - b)$ is beaten trivially, and P_1 certainly slides off a smooth table. If $b = d$, the bound becomes 1, so with $\mu < 1$ it always falls off, which is right because P_1 is already at the edge, still moving, when P_2 lands.
- Since $\mu < 1$, the second condition needs $e > \frac{1}{2}$. A nearly dead floor gives a short hop, so P_2 lands again first, as we'd expect.
- Numerically, with $d = 2$, $b = 1$, $\mu = 0.5$ and $e = 0.8$ (taking $g = 9.8$), stepping the motion gives $t_1 \approx 0.452\,\text{s}$ and $t_2 \approx 0.361\,\text{s}$, matching $V/(\mu g)$ and $2eV/g$ with $V^2 = 4.9$. Here $e = 0.8 < 1/(2\mu) = 1$, so P_2 bounces again first. Random tests of both criteria in a step-by-step simulation agreed in every case.

Common slips

- Keeping the tension in the second stage: once P_2 lands, the string is slack and P_1 is slowed by friction alone.
- Forgetting that P_1 has already moved a distance b, and comparing s with d rather than $d - b$.
- Writing the system's acceleration as $g(1 - \mu)$, forgetting that the force $mg(1 - \mu)$ has to accelerate *both* masses.

Beyond STEP

The string here is a *unilateral* constraint: it can pull but not push, so the equations of motion switch when it goes slack. Systems whose governing equations change at impacts or when contact is lost are called piecewise-smooth or hybrid systems, and they are studied seriously in engineering dynamics and robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats strings, friction and collisions carefully and is a good next step.

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